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		ACCEPTANCE TEST PROCEDURE		DOC CODE ATP	PART NUMBER 4556 111 101 38		
SWITCH FILTER ASSEMBLY (RFU R118)				ISSUE	SH No.	PREPD.	DATE
		04	5241649	04	2	ANIL KUMAR	22.11.13
		00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
		ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13
LIST OF ABBREVIATIONS: ATR ATTEN BPF COC CW DLVA EQUI FOD MDI PP PSD SOLT SW VSWR Acceptance Test Report Attenuation Band Pass Filter Certificate Of Conformance Continuous Wave Detector Logarithmic Video Amplifier Equaliser Foreign Object Debris Master Drawing Index Purchase Part Power Spectral Density Short Open Load Through SWitch Voltage Standing Wave Ratio							

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	00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
	ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13
1.0 RELATED DOCUMENTS: PP Drawing number: 4556 111 101 38. 2.0 FUNCTIONAL DESCRIPTION: Switch Filter Assembly (SFA) / Receiver Front-end Unit (RFU) operate in the frequency range of 1 - 18 GHz. It provides accurate amplitude information of the received RF pulses. The unit has got two input ports for RF IN and BIT. The unit has got an RF output port. The unit houses a 25 PIN D-type connector thru which power supply and control signals are fed to the unit and Log VID OUT is taken out. The functional block diagram is shown in Fig.1 Front end RF pre-amplification is provided to achieve a high sensitivity of -65dBm for amplitude measurement. The objective of SFA is to provide accurate amplitude information on the received RF pulses. In addition, a RF output is available with a nominal RF gain 35 ± 3 dB for further processing. 2.1 BRIEF DESCRIPTION OF SFA A20-MH166 is an RF unit in the frequency range of 1-18 GHz with a dynamic range of 0dBm to -65dBm. The RF Unit is a front-end receiver, which performs amplification and detection of RF signals received from the antenna and provides RF as well as video outputs to the Signal processor Unit (SPU) for further processing. The frequency band of 1-18 GHz is subdivided into 5 sub-bands for reception. The RF Unit receives signals in one of the five sub bands as per band control input from the Signal Processor. The received signal is amplified, passed through a switched filter bank and then processed by an Extended Range Detector and Log Video Amplifier (XDLVA) to obtain required dynamic range. It also gives an RF output at the IFM port for frequency measurement. The signal processor controls the RF sub-band path selection. Block diagram of the unit is given in Fig 1. Main features of this unit are given below. • A separate Band pass filter is included in the path for 1-2 GHz signal to suppress harmonic signal in this band, which can cause problems in integration. • Band switching time is specified to be 300ns (Max). • Log slope is 25mV/dB • RF Unit protection logic contains a limiter at the input to limit the input power to +30dBm (CW). • Output is DC coupled • Sealed unit with hermetic connector						



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		04	5241649				
		00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
		ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13
3.0 SPECIFICATIONS:							
3.1 ELECTRICAL SPECIFICATIONS:							
Sl. No.	PARAMETER			SPEC			
1.1	Frequency range			1 to 18 GHz covered in 5 Bands: 1- 2 GHz,2-6 GHz,6-10 GHz,10-14 GHz, 14-18 GHz			
1.2	Log dynamic range						
1.2.1	Nominal			-65dBm to 0dBm			
1.2.2	Log range where accuracy is defined			-65dBm to 0dbm with ±3dB accuracy			
1.2.3	Log reference voltage at -65dBm			250mV			
1.3	Absolute accuracy of power measurement in the centre of sub bands and 25°C ambient at the mounting surface, pass band around a straight line with co-ordinates 250mV @ -65dBm and 1875mV@0dBm with a slope of 25mV/dB			±2.0dB			
1.3.1	At any frequency or temperature around the line with co-ordinates -65dBm @ 250mV 0dBm @ 1875mV			±3.0dB			
1.4	Channel to channel Isolation for any frequency in the pass band, for an input such that the RF output is not compressed			65dB as seen on the Spectrum analyzer			
1.5	Antenna port to BIT Port Isolation			65dB			
1.6	Absolute maximum Input level						
1.6.1	For RF input Port			+30 dBm CW +50 dBm peak with 1µs PW @1000µs PRF			
1.7	Output RF port						
1.7.1	Small signal gain within pass band			35dB ±3.0dB			
1.7.2	Output SNR at -65dBm input			2.0 dB Min			
1.7.3	Harmonics(In Band)			-8dBc at saturation -30dBc in linear range			
1.7.4	Band Switching time Input TTL to 90% of selected Band			300ns Max			
1.8	RF input pulse width			100 ns to 14 µs			
1.9	Video port			Defined for an equivalent single ended output. RFU source impedance:93Ω Cable impedance:93Ω Load impedance:93Ω			
1.9.1	Output DC offset			25mV ± 25 mV			

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Sl. No.	PARAMETER	SPEC					
1.9.2	Output coupling	DC					
1.10	Video output (peak relative to ground)						
1.10.1	At -65dBm	250 mV ± 75 mV					
	At 0dBm	1875±75mV					
	When RFU is terminated into 93Ω±1Ω by 1meter (Max) long 93Ω cable						
1.10.2	Log slope	25 mV/dB					
1.10.3	Log linearity at a given frequency and temperature at 25mV/dB	±1.5dB					
1.11	Pulse characteristics	With properly terminated source impedance & load impedance for the line driver & line receiver					
1.11.1	Rise time (10% to 90%)	25 ns Max					
1.11.2	Settling time (10% to 0.5 dB below the final value)	40 ns Max					
1.11.3	Recovery Time	75 ns Max					
1.11.4	Falling Edge re-entrance for the pulse width 100ns to 100µs	The positive over shoot, excursion and ringing at the end of the falling end shall remain under threshold 110mV					
1.12	Switching spikes	Should be less than 110mV					
1.13	Delay between 50% TTL enable signal to 3dB(75mV) below peak of video shall be	Max 210 ± 20ns					
1.14	Duty cycle variation for maintaining the peak of pulsed signal relative to ground within the accuracy specification of Para 1.3	100%					
1.15	DC power +12V @ 540mA -12V @ 330mA +8V @ 830mA	17 Watts (Max)					
1.16	Connectors details						
1.16.1	Antenna Port	SMA-F					
1.16.2	BITE Port	SMA-F					
1.16.3	Receive Disable (Internal) (Receiver input termination 50Ω)	Available					
1.16.4	RF output Port	SMA-F					
1.16.5	DC power & Signal connector	25 Pin D-Sub(M) Hermetic Seal Corp P/N SR50511-3-25P-NK21					

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SI NO.	PARAMETER	SPEC			
1.16.6	Connector Pin details	Pin No.	Description	Pin No.	Description
		1	+12V DC	14	A1
		2	-12V DC	15	A0
		3	NC	16	A2
		4	/A0	17	NC
		5	/A1	18	+8V RTN
		6	/A2	19	NC
		7	/MS0 (RF EN/DIS)	20	NC
		8	NC	21	MS1(ANT/BITE)
		9	+8V	22	/MS1(ANT/BITE)
		10	NC	23	VID OUT
		11	+12V RTN	24	VID RTN
		12	-12V RTN	25	MS0(RF EN/DIS)
		13	NC		
1.17	Band selection control logic	A2	A1	A0	Selected Band
		0	0	1	1-2GHz
		0	1	0	2-6GHz
		0	1	1	6-10GHz (Default)
		1	0	0	10-14GHz
		1	0	1	14-18GHz
1.18	Antenna, BITE & Receiver Disable control	MS0	MS1	Mode	
		0	0	BIT mode RF	
		0	1	ANT mode RF(Default)	
		1	0	Receiver Termination to 50Ω	
		1	1	Unused	

3.2 MECHANICAL SPECIFICATIONS:

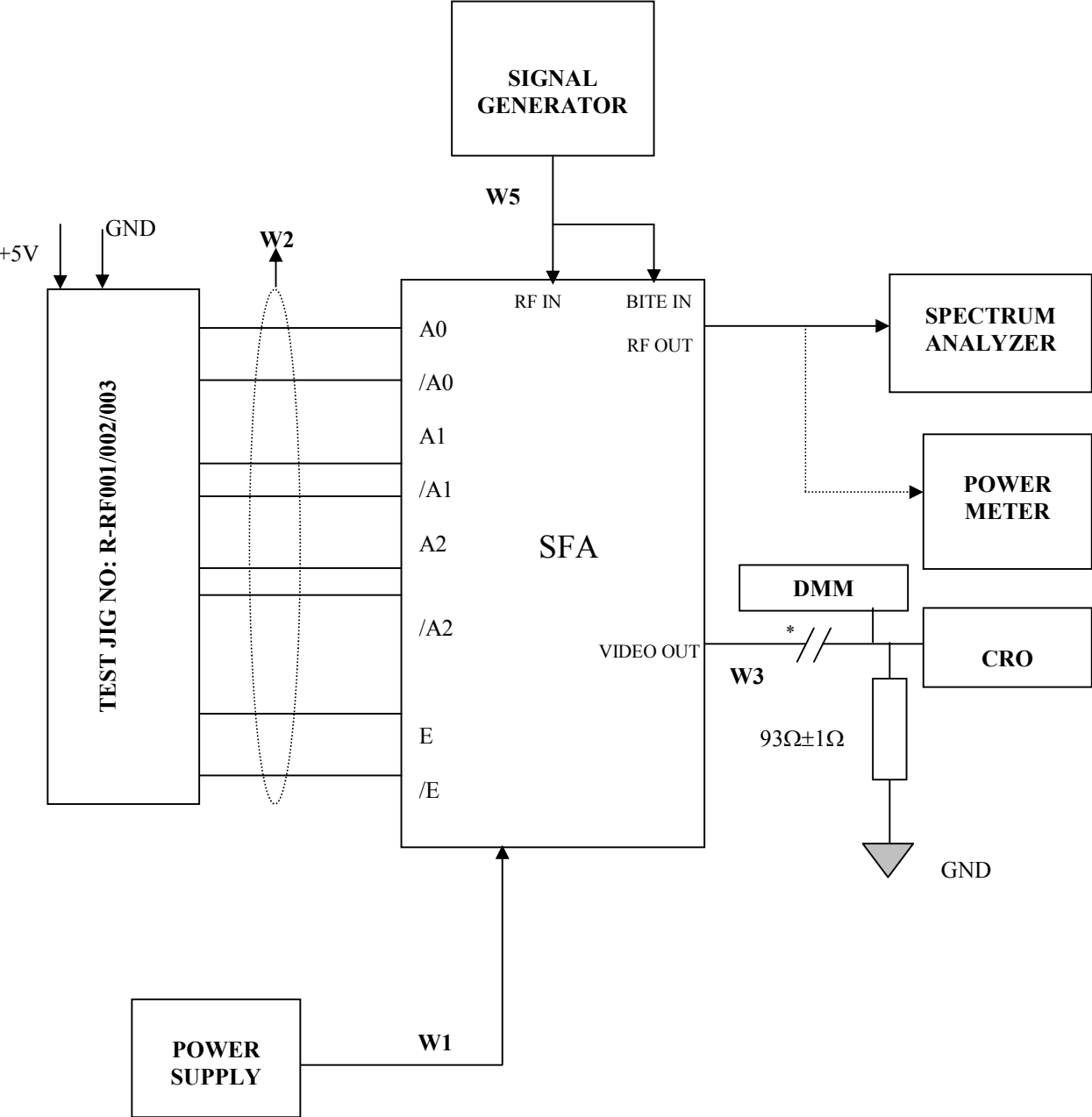
SIZE : 199.39mm X 88.77mm X 31.24mm

WEIGHT: 1.4 Kg max

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	ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13						
4.0 TEST PROCEDURE: 4.1 PHYSICAL INSPECTION: Verify requirements as per the below list and enter the results in Test report 7.1 on the offered unit. Physical inspection of the unit to be done as per BEL drawing no: 4556 111 101 38 PP as per latest configuration control document when required. - Details of RFU R118 viz. SI No, Year code and BEL P/N. - QM/EW&A accepted report of the Unit. - OEM ATR for RFU R118. - Verify for FOD – solder splash, Paint or lose items etc. - Verify overall dimensions of RFU R118 as per PP drawing. - Verify dimensions and centre distance(CD) of the mounting holes of RFU R118 with Fixture no: 8137 195 300 95 - Verify weight of RFU R118. - Verify COC of the OEM. 4.2 TEST INSTRUMENTS AND ACCESSORIES: 1. Multi output Power supply (+12V, -12V, +8V) <table><tr><td>+12V</td><td>1A</td></tr><tr><td>-12V</td><td>0.5A</td></tr><tr><td>+8V</td><td>1A</td></tr></table> -1 no. 2. Oscilloscope 500 MHz, Tektronix TDS 3052B or equivalent -1 no. 3. Signal generator, Agilent E8257D or equivalent -1 no. 4. Power meter, Agilent E4416A or equivalent -1 no. 5. Spectrum Analyzer, Agilent E4407B or equivalent -1 no. 6. Low Loss RF Cable -2 no. 7. Vector Network Analyzer, Agilent E8362B or equivalent -1 no. 8. Digital Multimeter, Agilent 34401A or equivalent -1 no. 9. 50 Ω Terminator, OR equivalent -1 no. 10. Cable, Video (93 Ohms) -1 meter. 11. Test Jig No: R-RF001/002/003 (Requires +5V external supply) -1 no.							+12V	1A	-12V	0.5A	+8V	1A
+12V	1A											
-12V	0.5A											
+8V	1A											

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			00ORIGINAL	No. OF SH 26	APPRD. DAMODARAN V	DATE 22.11.13
			ISS No. CN No.	HELD IN CD-1		

4.3 TEST SETUP



* 1 Meter Video Cable

Fig 2: RFU Test Set Up.

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				00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
				ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13

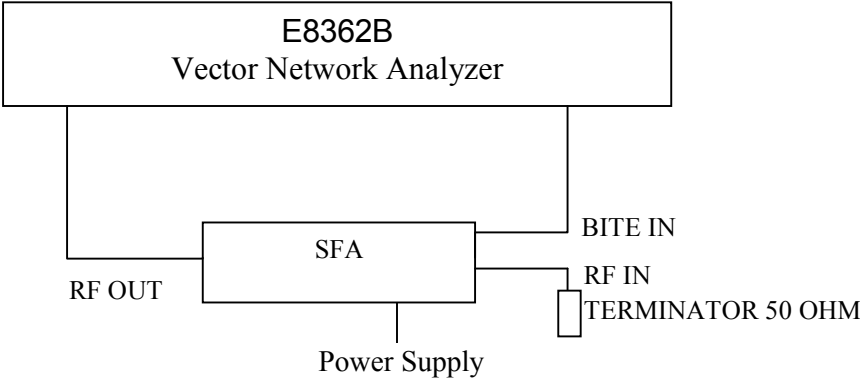


Fig 3: Test set-up for Gain Measurement

4.3.1 CONNECTION DETAILS:

LOOM	SIGNAL NAME POWER SUPPLY	SFA CONNECTOR - J2 PIN NO.	Type of Wire
W1	+12V	1	7/0.2 WIRE
	-12V	2	
	+8V	9	
	GND	11,12	

LOOM	CONTROL SIGNALS	SFA CONNECTOR - J2 PIN NO.	Type of Wire
W2	A0	15	7/0.2 WIRE
	A1	14	
	A2	16	
	/A0	4	
	/A1	5	
	/A2	6	
	/MS0	7	
	MS0	25	
	MS1	21 (ANT/BITE)	
	/MS1	22 (ANT/BITE)	

LOOM	Signal Name	SFA CONNECTOR- J2 PIN NO	Video Cable P/N
W3	VID OUT	23	93 Ohms cable
	VID RTN	24	93 Ohms cable

LOOM	Signal Name	SFA CONNECTOR	Type of Wire
W4	RF OUT	J5	Low loss RF cable
W5	RF IN	J4	Low loss RF cable
W6	BITE IN	J3	Low loss RF cable

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4.4 FUNCTIONAL TEST PROCEDURE:

Note: Terminate all unused ports with 50 Ohms terminator.

4.4.1 POWER CONSUMPTION MEASUREMENT:

SUPPLY VOLTAGE	MAXIMUM CURRENT LIMIT
+ 12 V	540 mA
- 12 V	330 mA
+8 V	830 mA

1. Make the inter connections as per the set-up shown in Fig 2.
2. Switch on the Power supply.
3. Check all the Voltages.
4. Measure the current in the +12 V, -12 V & +8 V points at quiescent conditions of SFA. Record the results in the test report SI.No 1.
5. The measured values should be as per the specification below.

4.4.2 RF MEASUREMENTS:

4.4.2.1 GAIN MEASUREMENT:

➤ PASS BAND FREQUENCIES

Spec: RF Gain: 35dB±3dB

1. Make the interconnections as per the set-up shown in the Fig 3.
2. Calibrate the Network Analyser System for Transmission measurement using “thru” calibration.
3. Attenuators may be used to get proper RF I/P to the SFA. Appropriate correction factors to be applied for the same.
4. Select Antenna port and Band 1 as per Table 1 & 2.
5. Feed RF I/P of -40dBm to RF I/P port of SFA in CW mode.
6. Make the Sweep measurements from 1GHz to 18GHz by selecting different bands

Band 1	: 1 to 2 GHz
Band 2	: 2 to 6 GHz
Band 3	: 6 to 10 GHz
Band 4	: 10 to 14 GHz.
Band 5	: 14 to 18GHz.

7. When the swept measurements are completed plot/print the band response.
8. Repeat the measurement for Band 2 to Band 5.
9. Record the test results in the test report SI.No 2
10. The measured values should be within the specification.

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	00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
	ISS No.	CN No.	CD-1	26	DAMODARAN V	22.11.13
Note: ◆ Make the interconnections as per set up shown in Fig: 2 for the following measurements from 6.2.2 to 6.2.7. ◆ Switch on the power supply. ◆ Measure the losses of RF cable connected between Signal Generator and I/P port of SFA at 1GHz, 1.5GHz, 2GHz, 4GHz, 6GHz, 8GHz, 10GHz, 12GHz, 14GHz, 16GHz & 18GHz. ◆ Apply this loss correction when feeding signal to SFA. ◆ Refer Table-1 &2 for selection of PORTS and Bands. 4.4.2.2 NOISE POWER MEASUREMENT Spec: Output Broad Band Noise: -30dBm Max. In all Bands 1. Terminate RF I/P and BITE ports with 50Ω. 2. Follow Table-1 for band selection and select Band-1. 3. Measure the O/P power level using Power Meter. 4. Record the results in Test Report. 5. Repeat above for Band-2, Band-3, Band-4 & Band-5 and record the measured values in the test report SI.No 3. 6. The measured values should be within the specifications shown above. 4.4.2.3 SNR MEASUREMENT Spec: Output SNR: 2dB Min in all Bands 1. Set the power level of input signal to -65 dBm & frequency to 1.5GHz. 2. Select antenna port and Band-1 as per Table-1 & Table-2. 3. Measure the power level of output signal (S) in dBm. Using Power Meter. 4. Switch off the input signal and measure the output power level. This gives the Noise level (N) in dBm. 5. The difference between S & N (S-N) gives SNR. 6. Repeat the same for 4GHz, 8GHz, 12GHz, 16GHz and record the measured values in the Test report SI.No 4. 7. The measured values should be within the specifications shown above. 4.4.2.4 HARMONICS MEASUREMENT Spec: Harmonics: Better than 8dBc. 1. Make the test set-up as shown in Fig 2, with Power Meter. 2. Select antenna port and Band-1 as per Table-1 & Table-2. 3. Set the frequency of input signal to 1.5 GHz and Power level to 0dBm. 4. Record the harmonics level in Test Report. 5. Repeat the above for other frequencies of 2GHz, 4GHz, 6GHz and 8GHz & record the measured values in the Test report SI.No 5. 6. The measured values should be within the specifications shown above.						

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4.4.3.2 RISE TIME MEASUREMENT

Spec: At band center: 1.5GHz, 4GHz, 8GHz, 12GHz, and 16GHz
Rise time ≤ 25 ns

1. Make the interconnections as per test set up shown in Fig 2.
2. Feed the pulse from signal Generator with following pulse parameters
PW = 1μs, PRI = 10 μs, Frequency = 1.5 GHz
3. Set input power level to -45dBm & input RF signal to Band-1 by following the Table-1.
4. Observe the Video Out on the CRO and measure the Rise Time. [During rise time measurements take the Sys. Signal Rise Time for correction].
5. Repeat the above measurements for Band-2, Band-3, Band-4 and Band 5 at band centers and record the results in the Test report SI.No 9.
6. The measured values shall be within the specifications

5.0 ESS TESTS: ESS test report supplied by the vendor will be verified by TS/EW.

SI NO.	PARAMETER	SPEC
4.1	Acceptance Test	All units shall undergo successful E.S.S test prior to acceptance test.
4.2		Visual examination and performance check of the unit
4.3	Stabilization Bake	Without applying any DC power, the unit would be at 125°C for 168Hrs. No data is required during test.
4.4		Visual examination and performance check of the unit
4.5	Thermal Shock	As per MIL-STD-202, Method 107 a. Temperature:-54°C and +125°C b. Dwell time at each temperature: 30minutes c. Transfer time from one temperature to the other: 15sec Max. d. The above constitutes 1-cycle. Conduct the Performance check at the end of 10 th and 20 th cycles, total 20 cycles.
4.6	Random Vibration	As per MIL-STD-883, Method 2026, test condition E, 16.4 grms from 50Hz to 2000Hz. Test for 10 minutes in the z-axis (perpendicular to the mounting axis) and 5 minutes in the other 2 axes. During vibration, the single point (12GHz and RF I/P=-50dBm) RF gain and transfer response would be mounted and recorded.
4.7		Conduct the performance check
4.8	Mechanical shock	As per MIL-STD-202F, Method 213B, 100g, 6ms, Half sine 5-shock cycles along y-1 axis
4.9		Conduct the performance check

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SI NO.	PARAMETER	SPEC																											
4.10	Temperature Cycling	The unit shall be subjected to 24 temperature cycles. Each temperature cycle is described below and takes 3.5Hrs																											
		Cycle description 1 st and 24 th cycles:																											
		<table><tr><th>Temp</th><th>Time</th><th>DC Power</th></tr><tr><td>-40°C Cold soak</td><td>30min</td><td>OFF</td></tr><tr><td>-40°C wake up & warm up</td><td>5min</td><td>ON</td></tr><tr><td>-40°C operational test</td><td>25min</td><td>ON</td></tr><tr><td>Transition to +71°C</td><td>15min</td><td>ON</td></tr><tr><td>Soak at +71°C</td><td>60mins</td><td>ON</td></tr><tr><td>+71°C operational test</td><td>25min</td><td>ON</td></tr><tr><td>Total dwell time at +71°C</td><td>2.5Hrs</td><td>ON</td></tr><tr><td>Transition to -40°C</td><td>15min Max</td><td>OFF</td></tr></table>	Temp	Time	DC Power	-40°C Cold soak	30min	OFF	-40°C wake up & warm up	5min	ON	-40°C operational test	25min	ON	Transition to +71°C	15min	ON	Soak at +71°C	60mins	ON	+71°C operational test	25min	ON	Total dwell time at +71°C	2.5Hrs	ON	Transition to -40°C	15min Max	OFF
		Temp	Time	DC Power																									
		-40°C Cold soak	30min	OFF																									
		-40°C wake up & warm up	5min	ON																									
		-40°C operational test	25min	ON																									
		Transition to +71°C	15min	ON																									
		Soak at +71°C	60mins	ON																									
		+71°C operational test	25min	ON																									
Total dwell time at +71°C	2.5Hrs	ON																											
Transition to -40°C	15min Max	OFF																											
Cycle description 2 nd through 23 rd cycles:																													
<table><tr><th>Temp</th><th>Time</th><th>DC power</th></tr><tr><td>-40°C cold soak</td><td>30min</td><td>OFF</td></tr><tr><td>Transition to +85°C</td><td>15min</td><td>ON</td></tr><tr><td>dwell time at +85°C</td><td>2.5Hrs</td><td>ON</td></tr><tr><td>Transition to -40°C</td><td>15min Max</td><td>OFF</td></tr></table>	Temp	Time	DC power	-40°C cold soak	30min	OFF	Transition to +85°C	15min	ON	dwell time at +85°C	2.5Hrs	ON	Transition to -40°C	15min Max	OFF														
Temp	Time	DC power																											
-40°C cold soak	30min	OFF																											
Transition to +85°C	15min	ON																											
dwell time at +85°C	2.5Hrs	ON																											
Transition to -40°C	15min Max	OFF																											
Conduct performance check at the end of the 6 th , 12th, 18th and 23 rd cycle. During the 1 st and 24 th cycle monitor the performance of the unit.																													
4.11	RF Burn-In:	With DC power ON, at 0dBm input, at +85°C for 48Hrs minimum.																											
4.12		Conduct the performance check.																											

6.0 COMPATIBILITY CHECK

RFU shall be checked in R118 system integration stage by giving frequencies 1.5GHz, 4.5GHz, 7.5GHz, 12GHz & 15GHz at power level of -30 dBm. If there is valid symbol as per PFM, the RFU is considered as compatible with R118 system.

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7.0 TEST REPORT:										
Tests Conducted for the Contract No / Sale order (Ref No): _____										
Drawing No: 455611110138			Serial No:			Date:				
7.1 PHYSICAL INSPECTION REMARKS:										
Activity				Observation		Acceptance by BEL Rep		Acceptance by ORDAQA Rep		
Initial physical inspection										
a) Check for over all dimensions (mm) (Tolerances as per PP drawing)		L	199.39							
		B	88.77							
		H	31.24							
b) Following dimension to be verified with Inspection Jig. Tool No.: 8137 195 300 95										
Reference Zone in drawing: 4556 111 101 38 PP		Dimensions in mm (Tolerance as per drawing 4556 111 101 38 PP)								
A2		82.93								
B1		98.55								
B2		88.14								
c) Following dimension to be verified with drawing 4556 111 101 38 PP										
Reference Zone in PP drawing		Dimensions in mm (Tolerance as per drawing)								
B1		3.3								
B3		23.62								
B3		7.19								
B3		9.87								
d) Check marking details as per drawing 4556 111 101 38 PP										
e) Visual inspection										
1.SMA female connector:		3 Nos								
2. D-type connector - 25 pin (As per MIL-C-24308):		1No								
3.Screen Print Texts (As per drawing 4556 111 101 38)										
f) Following to be verified with respect to CoC.										
1. Paint 37038 texture Black per MIL-C-85285										

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		BAND 4	NA																																																										
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9. RISE TIME MEASUREMENT (Para 4.4.3.2) Specified Value: ≤ 25nS Max. <table><thead><tr><td></td><td>@ PW= 1μS,PRI=10μS</td><td>Test Result (Pass/ Fail)</td><td>Remarks</td></tr></thead><tbody><tr><td rowspan="2">BAND</td><td>RISE TIME (nS)</td><td rowspan="2"></td><td rowspan="2"></td></tr><tr><td>@ -45dBm I/P</td></tr><tr><td>BAND1</td><td></td><td></td><td></td></tr><tr><td>BAND 2</td><td></td><td></td><td></td></tr><tr><td>BAND 3</td><td></td><td></td><td></td></tr><tr><td>BAND 4</td><td></td><td></td><td></td></tr></tbody></table> BEL Rep Date: OAQA Rep Date:									@ PW= 1μS,PRI=10μS	Test Result (Pass/ Fail)	Remarks	BAND	RISE TIME (nS)			@ -45dBm I/P	BAND1				BAND 2				BAND 3				BAND 4			
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ANNEXURE - A

A. PRODUCTION QUALITY TESTING (PQT)

A1. Periodicity of PQT: PQT shall be conducted for RFU during the first quarter of the year, for 1 in every 40 units.

A2. Conditions for PQT: PQT shall also be conducted for the RFU under the following conditions,

- a. Changes in Design/Hardware.
- b. Change of Material.
- c. Change of Manufacturer.
- d. Change in Place of manufacture.

A3. Screening for PQT: PQT to be conducted only for ATP and ESS qualified units.

A4. Environmental Tests:

The following environmental tests will be carried out on SFA.

- 1. Random Vibration Test
- 2. Low Temperature Storage & Operational Test
- 3. High Temperature Storage & Operational Test
- 4. Humidity Test
- 5. Altitude test

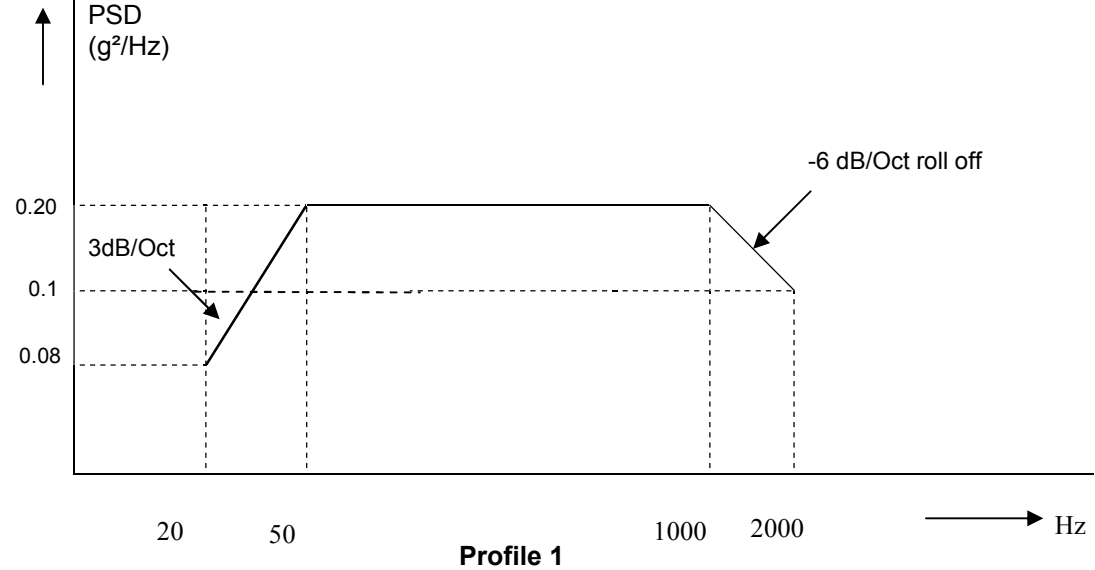
Prior-to & after each environmental test all ATP tests, mentioned in this document, are to be carried out.

Note:

After completion of Low & High temperature storage tests, the unit shall be brought to ambient temperature and allowed to stabilize for 2 Hrs before starting operational test.

A4.1 Random vibration: (Operational)

Random Vibration with power spectral density from 20 Hz to 50Hz rising to 0.2g²/Hz with 3dB/Octave, 50Hz to1000Hz with 0.2g²/Hz & -6dB/Octave roll off from 1000Hz to 2000Hz. 15 Minutes on all three axes. Refer Profile 1.



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A4.1.1 OPERATIONAL TESTING													
Transfer Response test as per table below is to be carried out.													
Frequency (GHz) Input Power(dBm)		Band 1		Band 2		Band 3		Band 4		Band 5		Spec: 1125 mV±75 mV	
		1.5		4		8		12		16		Test Result Pass/Fail	Remark s
		Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)		
-30dBm													
A 4.2 Low Temperature- Storage & Operation													
A4.2.1 Storage													
SFA unit shall be subjected to Low Temperature Storage Test in accordance With the following parameters and Profile.													
Chamber Temperature : -54°C. Duration : 4 Hrs.													
A4.2.2 Operating													
SFA shall be subjected to a Low Temperature Operational Test in accordance with the following parameters.													
Chamber Temperature : -40 °C Duration : 4 Hours Continuously ON													
A4.2.2.1 OPERATIONAL TESTING													
Transfer Response test as per Para A4.1.1 to be carried out.													
A4.3 High Temperature - Storage & Operation													
A4.3.1 Storage													
SFA unit shall be subjected to High Temperature Storage Test in accordance with the following parameters and Profile.													
Chamber Temperature : +120°C Duration : 12 Hrs.													

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A4.3.2 Operation

SFA shall be subjected to a High Temperature Operational Test in accordance with the following parameters.

Chamber Temperature : +71 °C
Duration : 4 Hours Continuously ON

Note: The amount of time required to ramp the temperature is dependent on the test facility change rate. The volume of the chamber shall be at least 10 times the DUT.A ground has to be extended from DUT to test setup.

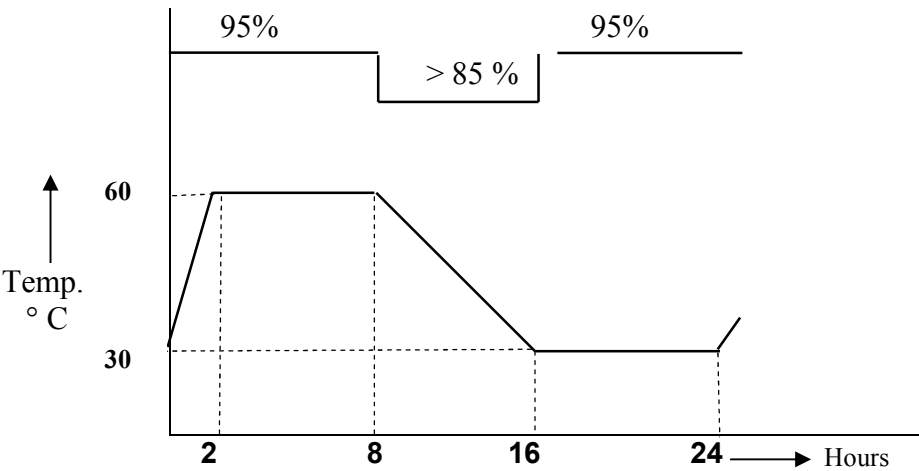
A4.3.2.1 OPERATIONAL TESTING

Transfer Response test as per Para A4.1.1 to be carried out.

A4.4 Humidity (Non Operational)

SFA shall be subjected to a Damp Heat (Humidity) Test consisting of 5 Cycles in accordance with the following parameters and Profile

Humidity : 85 to 95% R.H
Temperature : +30 to +60 ° C
Cycle Period : 24 Hrs.
No.of Cycles : 5



DURATION: 5 Cycles

Note:

1. The amount of time required to ramp the temperature is dependent on the test facility change rate.
2. At the end unit shall undergo Visual Inspection

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A4.5 Altitude (Operational) SFA shall be subjected to altitude test consisting of 1 Cycle in accordance with the following parameters as per MIL-STD-810F, Method 500.4 Altitude : 20Km Duration : 1hour soak Altitude change rate : 10 Meter/Sec (can be changed subject to chamber conditions) A4.5.1 OPERATIONAL TESTING Transfer Response test as per Para A4.1.1 to be carried out. Table-1 (PQT) <table><tr><th>Sl</th><th>Test</th><th>Compliance (Y/N)</th><th>Signature (BEL Rep)</th><th>Signature (AQA)</th><th>Remarks</th></tr><tr><td>1</td><td>Random Vibration Test</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>Low temperature-Storage</td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>Low temperature-Operational</td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>High temperature-Storage</td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>High temperature-Operational</td><td></td><td></td><td></td><td></td></tr><tr><td>6</td><td>Humidity (Non Operational)</td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td>Altitude (Operational)</td><td></td><td></td><td></td><td></td></tr></table>								Sl	Test	Compliance (Y/N)	Signature (BEL Rep)	Signature (AQA)	Remarks	1	Random Vibration Test					2	Low temperature-Storage					3	Low temperature-Operational					4	High temperature-Storage					5	High temperature-Operational					6	Humidity (Non Operational)					7	Altitude (Operational)				
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 भारत इलेक्ट्रॉनिक्स <i>BHARAT ELECTRONICS</i>		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 111 101 38			
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ANNEXURE - B

